

Understanding Technical Interviews

DAY 96
14/10/2025

Why this technical interview process exist?

candidate might have a question in mind He/she is good at some tech stack (Application development, AI or any tech stack) But what the relation with the Data Structures.

In Technical Interview Data structure Based Questions are asked

You might heard people saying Solve 500 problems or solve 300 problems,

few says DSA problems solving require pattern recognition

Someone created list of 30 patterns Some one created some list which have DSA Questions.

Some say Count doesn't matter Solve 75 problems

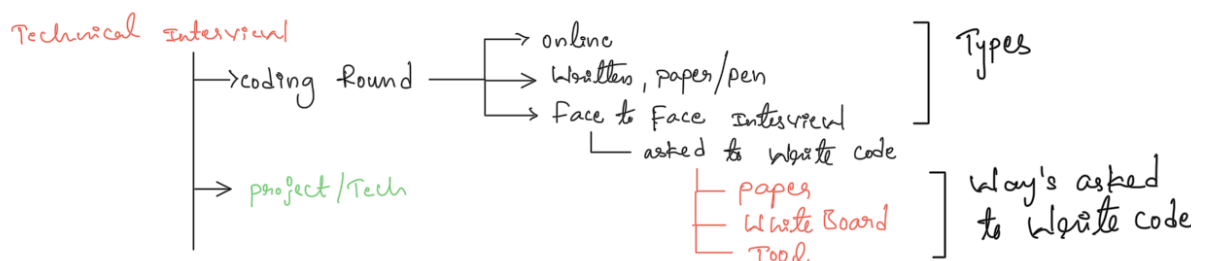
There are so many youtube channels saying different insights to crack technical interview.

But what we think in the perspective of other side person in the interview, The person is interviewer.

How the interviewer judge candidate & what skill they expect from the candidate's in the technical interview.

What skill & competency the candidate should demonstrate?

Before that understand what does technical interview in Big tech companies consists of?



The Companies Really Look For

- ① Problem Solving Skill
- ② Coding
- ③ Application of CS fundamentals
- ④ Communication
- ⑤ Attitude

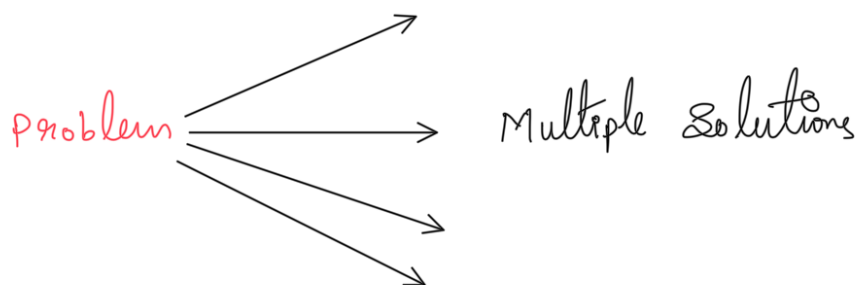
→ How problem Solving of Candidate is tested?

In Software Industry The Candidate should have Ability of understanding situation, debugging, creative thinking, logical thinking, Very calm and composed approach to look at problems and finding Solutions

Most Cases the technical problem given in a interview Have Multiple Solutions the interviewer analyse the Candidate Based on How the Candidate do the trade off's between Solutions. and Which Solution is used in What Scenario.

Interviewer Have to go Non Judge you on this

To Judge the Candidate's problem Solving Skill The Interviewer ask a Question Which have Multiple possible Solution, and See How Candidate comes up With Solutions and their Trade off's.



Tomorrow I will dig deeper into how technical interview conducted and what the company expect from the candidate

🧩 1. Why does the technical interview process exist?

Purpose:

Technical interviews exist to **evaluate how well a candidate can apply their technical knowledge to real-world problems**.

It helps companies ensure that a person can not only "know" things but also **think logically, debug issues, write clean code, and build reliable systems** under pressure.

Main reasons:

- To verify the candidate's **core computer science fundamentals** (DSA, OOP, DBMS, OS, etc.)
- To test **problem-solving ability** — how you approach, analyze, and break down a complex problem.
- To check **practical coding skills** — can you translate logic into working, efficient code?
- To assess **communication and collaboration** — how you explain technical ideas and reasoning.
- To see if you **fit into the company's engineering culture** — curiosity, learning attitude, ownership, etc.

⚙️ 2. What skills and competencies should the candidate demonstrate?

A strong technical candidate demonstrates a **blend of technical, analytical, and interpersonal competencies**:

Category	Competency	What Interviewer Looks For
Technical Fundamentals	Data structures, algorithms, OOP, DBMS, OS, networks	Accuracy, understanding, clarity of concepts
Problem Solving	Logical reasoning, analytical thinking, debugging	Structured approach, clarity in assumptions, edge cases
Coding Proficiency	Writing efficient, clean, readable code	Use of good variable names, proper indentation, correct syntax
System Design	Building scalable and maintainable systems	Modular thinking, scalability, trade-off analysis
Communication	Explaining ideas clearly and confidently	Ability to explain code, thought process, and choices
Attitude	Curiosity, humility, willingness to learn	Coachability, ownership, enthusiasm

3. The skills companies really look for

Top companies (like MAANG or startups) often focus on **these core areas**:

1. **Strong fundamentals** – DSA, OOP, DBMS, OS, and Networking.
2. **Analytical thinking** – how you analyze and simplify complex problems.
3. **Coding ability** – writing correct and efficient code under time constraints.
4. **Design thinking** – ability to design APIs, systems, or architectures.
5. **Soft skills** – teamwork, communication, adaptability.
6. **Learning mindset** – ability to pick up new tools, languages, or frameworks quickly.
7. **Ownership** – taking responsibility for your work and thinking beyond your role.

4. How problem-solving of a candidate is tested

Problem-solving is tested through **structured technical challenges** like:

Method	Example	What it Reveals
DSA coding problems	Arrays, recursion, dynamic programming	Logic, pattern recognition, optimization
Debugging tasks	Given buggy code to fix	Analytical & debugging ability
System design	"Design a chat app" or "URL shortener"	Abstraction, scalability, modularity
Real-world scenarios	"How would you optimize this API?"	Practical engineering approach
Take-home assignments / pair programming	Small project	Coding habits, collaboration, code quality

5. How interviewers judge a candidate's problem-solving skill

They don't just care about *the final answer* — they care about *how you think*.

Interviewers look for:

1. **Structured Thinking** – Breaking the problem into smaller parts.
E.g. "First, I'll check input constraints, then think about possible edge cases..."
2. **Clarity of Thought** – How clearly you explain your reasoning and trade-offs.
3. **Step-by-Step Approach** – Moving logically from idea → pseudocode → implementation.
4. **Optimization Skills** – Can you improve from $O(n^2)$ to $O(n \log n)$? Do you know when to stop optimizing?
5. **Error Handling & Edge Cases** – Do you test for nulls, empty inputs, limits?
6. **Communication** – Do you talk through your approach, ask clarifying questions, and justify decisions?
7. **Composure** – How you react under pressure or when stuck — calm reasoning vs. panic.

Summary

Aspect	Description
Why exists	To measure if you can apply technical knowledge in real-world settings
What to demonstrate	Core technical + problem-solving + communication skills
Skills companies value	DSA, design, clarity, collaboration, curiosity
How problem-solving is tested	Through DSA problems, debugging, and design challenges
How judged	On reasoning process, clarity, efficiency, and composure